

TABLE II
Voice Database and Sample Files

Database created after recording

Voice sample	Picture
OBJ1.wav	OBJ1.JPG
OBJ2.wav	OBJ2.JPG
OBJ3.wav	OBJ3.JPG
OBJ4.wav	OBJ4.JPG
OBJ5.wav	OBJ5.JPG

Run time recorded voice for recognition

Record.wav	—
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The user should press Object while speaking in front of the microphone. During this time, the program also captures the image of the user from the webcam and displays it on the screen. All five voice data are stored in the database along with image files as listed in Table II. After the database generation, you are ready to go!

Now, you can check the recorded voice samples by clicking Record. After successful voice recognition, the program displays the detected object and its sound properties. It generates the following plots of speech signal:

1. Waveform of recorded signal
2. Magnitude curve of recorded signal
3. Spectrogram of recorded signal
4. Cepstrum analysis of recorded signal
5. Curve of Fourier coefficient in complex plane

Peripheral and property section provides information about the connected peripherals like microphone and webcam.

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source. efymag.com

It also provides information like sampling rate of recorded speech signal, number of bits for encoding, channel information (dual/mono), pitch of the recorded speech signal and peak value in Cepstrum analysis. For any problems during recording your voice and photo, always check this section.

Steps for testing

1. Copy the complete folder (update) of this project from EFY DVD or website to your system.

2. Run demo.m program.

3. Type password 123456

4. Click on 'peripheral+property' option and then click on microphone and webcam to make sure that these are connected.

5. Face the camera and press Object 1 and speak a word, say, 'one,' near the microphone. It will record your voice sample as well as your photo. Wait until the capturing process finishes. Similarly, say 'two' for Object 2.

6. Repeat for Object 3 through Object 5 to prepare the database.

7. Now, give the run-time sample by clicking Record. Face the webcam and say 'two' near the microphone. It should match one of the five samples recorded earlier. Wait for some time till an 'object detected successfully' message comes on the screen (refer Fig. 4).

8. Press Recognition. It will give detailed information like different plots of the recognised voice sample (refer Fig. 5). If a match is not found, nothing will be displayed in the object detected box just below the captured image.

9. We can check the matching coefficient by clicking Error Comparison.

Future application

The program can be modified to interface with embedded systems through parallel or serial port of the PC for monitoring on LCD/LED displays. If a match is found, you can enable a lock or other external devices. ●

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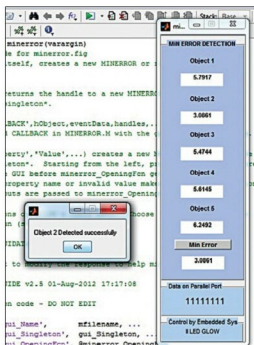


Fig. 4: Prompt to confirm object detection

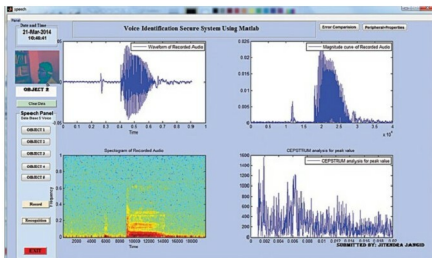


Fig. 5: Plots of the recognised voice sample